BidPilot: What We're Building

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The problem

Online sellers spend money on ads for each product and each search keyword. Someone has to check every day which ones make money and which ones waste it, then raise or lower the bids. Big companies hire bid managers for this. Small sellers can't afford one, so their bids stay wrong for weeks and money leaks out.

What we're building

An AI bid manager and consultant for product and keyword ads. It does the daily work a human bid manager does: reads the numbers, decides new bids, explains why, and asks a person before any big change. Once a week it also works as a consultant: it digs into the data, researches the market online, and gives advice with sources anyone can check.

We only cover two kinds of bids:

- **Product bids:** how much we pay when a specific product's ad is shown (like Google Shopping ads).
- **Keyword bids:** how much we pay when someone searches a specific word or phrase (like Google search ads).

Four ideas make it different:

1. **It knows the stock.** It won't spend money advertising a product that's about to run out, and it pushes products with too much stock.
2. **Math picks the numbers, AI explains them.** A calculator-style optimizer sets the bids. The AI decides the strategy, checks the rules, and explains every change in plain words with the numbers behind it.
3. **It never touches real money.** Everything runs in a practice ad account (the sandbox), so we can test safely and show results.

4. Every claim has a source you can click. A number links to the data table and the code that produced it, marked [D1], [D2]. A market fact links to the web page and the exact quote it came from, marked [W1], [W2]. Anything that can't be matched to a real source gets removed before anyone sees it.

Daily bidding workflows

These are the jobs a bid manager repeats every day. They run automatically every simulated day.

Workflow	What it does	Example of what the agent says
1. Daily check-up	Reads yesterday's results for every product and keyword and flags what changed	"Keyword 'cotton bedsheet' cost went up 40% but sales stayed flat."
2. Bid adjustment	Raises bids on things that make profit, lowers bids on things that lose money	"Raise 'king size cover' bid from Rs 18 to Rs 21. It earns Rs 4.10 for every Rs 1 spent."
3. Waste cleanup	Finds keywords and products spending money with no sales, and lowers or pauses them	"Pause 'free bedsheet'. Spent Rs 2,300 over 9 days, 0 sales."
4. Stock guard	Lowers bids on products about to sell out, pushes products with too much stock	"Product P-14 has 3 days of stock left. Lowering its bid 50% until restock."

The example numbers are made up to show the style. Real numbers come from our dataset.

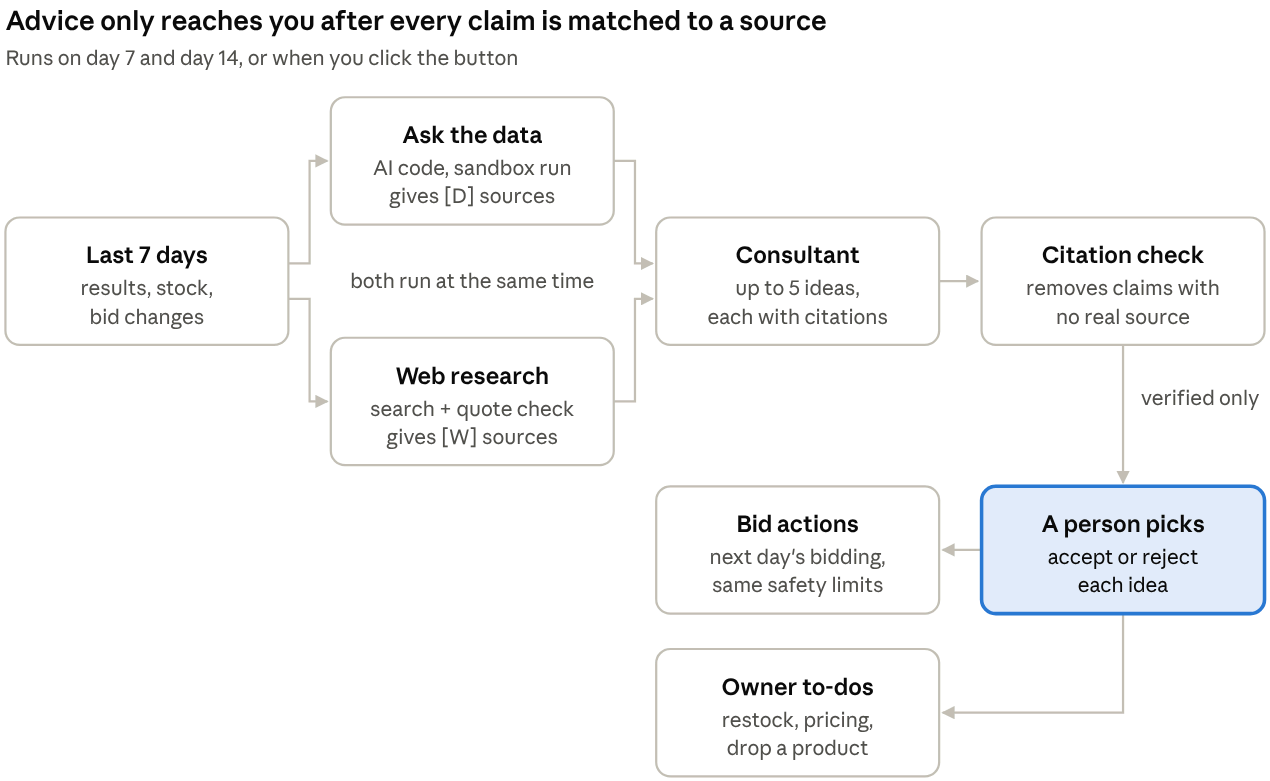
Weekly consultancy workflows

These answer the bigger questions that daily bid changes can't fix. They run on day 7 and day 14, or on request.

Workflow	What it does	Example of what the agent says
5. Ask the data	The AI writes a short piece of code to answer a question, runs it in a locked-down sandbox, and saves the result table as a source	"Keywords with 'free' or 'cheap' spent Rs 9,800 with zero sales [D1]."
6. Web research	Searches the web for market facts and keeps only facts whose exact quote is found on the page	"Bedding demand usually rises before winter [W2]."

Workflow	What it does	Example of what the agent says
7. Consultant report	Combines data and research into up to 5 recommendations, each with sources. The human accepts or rejects each one	"Stop targeting 'free'/'cheap' searches [D1]. Restock P-14 earlier: it was out of stock 5 days while ads kept running [D3]."

As before, the example numbers and facts only show the style. Real ones come from the data and real web pages.

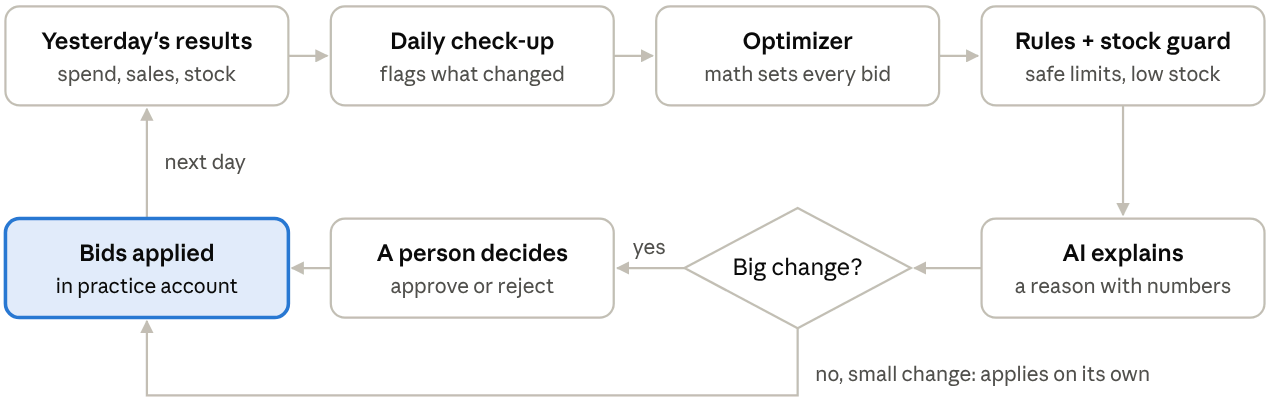


weekly consultant · analytics and research in parallel, then a citation check

Accepted bid ideas go into the next day's bidding and still pass the same safety limits. Ideas the agent can't do itself, like restocking or changing a price, become a to-do list for the owner.

How one day works

Every day, math sets the bids and a person approves only the big changes



one simulated day · 7 steps, 1 decision

A "big change" means a bid moving more than 10%, a pause, or any change on one of the top 10 spenders. In the 14-day fast-forward, approvals are auto-accepted so the chart can run, and we show one manual approval live.

What's real and what's simulated

Judges respect honesty, so we say this clearly in the pitch.

Part	Real or simulated	Why
The AI agent, optimizer, rules, approval step	**Real**	This is the product we built
AI models running on AMD GPUs	**Real**	We measure real speed on real hardware
AI-written analysis code running in the sandbox	**Real**	The code and its result table are shown for every [D] source
Web research and citations	**Real**	Real web pages with real links. We record the search once and replay it in the demo so a bad internet connection can't break it. The retrieval date is shown

Part	Real or simulated	Why
The ad account	**Simulated**	Google and Meta take days to approve ad API access, and we won't risk real money
Products, keywords, stock and ad results	**Simulated**	We generate a realistic fake store with 40 products and about 120 keywords
How the market reacts to bid changes	**Simulated**	A hidden formula decides results. The agent never sees it and has to learn from the data, like a real bid manager

One line for judges: "The ad account is a practice account. Everything the agent does would work the same way with a real Google Ads connection, which is our next step."

What we are NOT building

If someone suggests one of these during the hackathon, the answer is "after the hackathon."

- Connecting to real Google Ads, Meta, or Daraz accounts
- Writing ad copy or images
- Social media posting
- Audience, location, or device bidding (only product and keyword bids)
- Open-ended chat with the agent ("Ask the data" answers questions about the data only)
- Login, user accounts, or a polished website. A simple dashboard is enough

Where AMD fits

We run open AI models ourselves on an AMD GPU in the cloud, using vLLM (the standard tool for serving models on AMD). Speed is our AMD story, and we prove it with a measured number, not a claim.

- **Slow way (what most people would build):** one big AI model looks at every keyword, one after another, and guesses each bid.

- **Our way:** the optimizer calculates all bids in under a second. A small, fast model writes all the explanations at the same time instead of one by one. The big model is called once a day for strategy.

The slow way is also less accurate, because AI models are bad at math. We do the same for the consultant: its data questions and web searches run at the same time instead of one after another. We show both comparisons side by side in the demo.

The demo (4 minutes)

The money shot is one chart: 14 simulated days, three lines for profit. Our agent should end clearly above the other two.

1. **Fixed bids:** nobody touches the bids. This is the lazy seller.
2. **Simple rules:** "if it loses money, lower 10%." This is what basic tools already do.
3. **Our agent:** optimizer + stock guard + AI explanations + human approval.

Demo script:

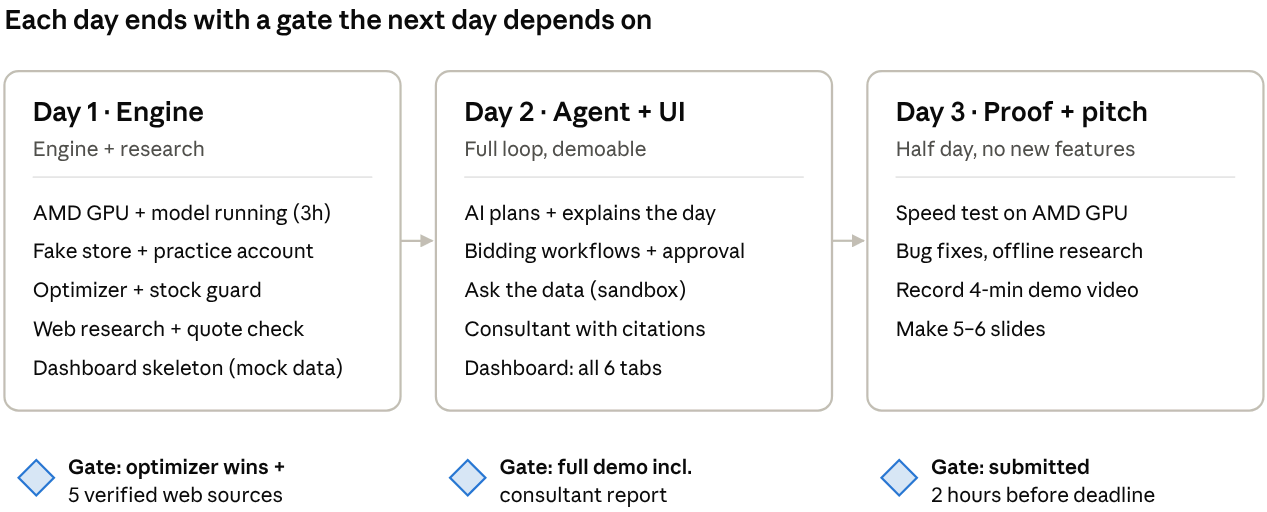
1. Show the fake store: 40 products, ~120 keywords, some running out of stock (20 sec).
2. Click **Run day**. The agent shows its plan: "23 changes, 4 need your approval." Read one reason, approve one change, reject one (40 sec).
3. Show the 14-day profit race chart (30 sec).
4. Open the **Consultant** report. Click a [D] citation to show the code and table behind it. Click a [W] citation to show the web page and quote. Accept one recommendation (60 sec).
5. **Ask the data** live with one preset question (30 sec).
6. Show the speed comparison on AMD (30 sec).
7. Close with the honest line about the practice account and the next step (10 sec).

We win if the agent clearly beats both other lines, every change has a clear reason, the speed number is real, and every consultant claim opens to a real source.

Team roles and timeline

This assumes 3 people. Adding consultancy, research, and analytics only fits because research starts on Day 1 and analytics runs in parallel on Day 2. With 2 people, use the cut list in PLAN.md: preset questions only for "Ask the data," and research from saved results only.

Person	Owns	PLAN.md phases
A: Data + engine + analytics	Fake store, practice ad account, metrics, optimizer, stock guard, "Ask the data" sandbox	1, 2, 3, 6
B: AI + AMD + consultant	Model on AMD GPU, AI client, prompts, bidding workflows, consultant, speed test	0, 4, 5, 8, 10
C: Research + dashboard + pitch	Web research with quote check, dashboard, video, slides	7, 9, 11



2.5-day build plan · 3 phases, 3 gates

If a gate isn't met on time, cut scope from the next day instead of pushing the gate.

Rules to protect the deadline

- **Get the AI model running on AMD in the first 3 hours.** If it fails by then, switch to a smaller model. If AMD is down, the code also works with any OpenAI-style API so we can keep building.
- **The product works without AI first.** If the optimizer beats the other lines with no AI, the core is done. The AI adds explanations, judgment, and the consultant on top.
- **Demoable by the end of Day 2,** even if it looks ugly.

- **Day 3 adds no new features.** It's only for speed tests, fixing bugs, the video, and slides. Switch research to saved results so the demo works without internet.
- **Use Claude Code one phase at a time** from PLAN.md. Test each phase before starting the next.
- **Submit at least 2 hours before the deadline.**